



Department of: Information Systems

الفترة الأولى

Second Semester Examination – Academic year 2025/2026

Subject

Digital Logic Design

Class

AI 1 CS1

FORM (A)

Date

Lecturer

Dr. Mhsen Hussein AL-Haidary

Time

2 hours

*Note: Indicate your answer by shading the correct option for each question on the answer sheet.

Chose the correct answer :

1	A NOR gate is an OR gate followed by a..... gate A- XNOR ☒ B- OR C- AND ☒ D- NOT
2	An XOR gate outputs 1 when all inputs are..... ☒ A- Different B- Ones ☒ C- Equal D- Ones
3	Boolean variables can take only two values,..... A- Any real numbers B- 1 and 2 ☒ C- 0 and 1 D- None of the above
4	Binary number system has a base of..... ☒ A- 10 B- 8 ☒ C- 2 D- 16
5	An SR flip-flop has..... inputs ☒ A- Set and Reset B- Shift and Rotate ☒ C- Serial and Register D- None of the above
6	A NOR gate is a combination of ☒ A- OR + NOT B- AND + NOT C- XOR + NOT D- None of the above
7	$A + 1 = \dots$ in Boolean algebra ☒ A- 0 B- A C- A' ☒ D- 1
8	A minterm contains every variable exactly..... ☒ A- Zero times ☒ B- Once C- Twice times D- Three times
9	Which gate is a basic gate ☒ A- OR gate ☒ B- NAND gate C- NOR gate D- XOR gate
10	Binary number system uses : ☒ A- 8 digits B- 6 digits ☒ C- 2 digits D- 10 digits
11	logic gate that performs logical addition is A- AND ☒ B- OR C- NAND D- XOR
12	The output of an AND gate is HIGH when: A- Any input is HIGH ☒ B- All inputs are HIGH C- All inputs are LOW D- One input is LOW
13	2-variable K-map contains A- 2 cells ☒ B- 4 cells C- 8 cells D- 16 cells

14	Outputs of a half adder are: A- Sum and Carry B- Product and Carry C- Sum only D- Input and Output
15	An encoder converts: A- Many inputs to binary code B- Binary to decimal C- Analog to digital D- Digital to analog
16	The third input of a full adder is: A- Borrow B- Carry-in C- Output D- Clock
17	The output of an XOR gate is HIGH (1) when A- Both inputs are the same B- Both inputs are LOW C- Inputs are different D- None of the above
18	Combinational circuits depend on A- Present inputs only B- Previous outputs only C- Clock only D- Memory only
19	The output of an OR gate is LOW when: A- One input is HIGH B- All inputs are HIGH C- All inputs are LOW D- Any input is LOW
20	NAND gate is a combination of A- AND + NOT B- OR + NOT C- XOR + NOT D- NOR + NOT
21	A..... shows all possible combinations of inputs and output. A- K Map B- Logic diagram C- Demorgans theorems D- Truth table
22	The Boolean expression $A + A'$ is equal to A- 1 B- 0 C- A D- 2A
23	How many cells does a 3-variables K map contain : A- 6 cells B- 8 cells C- 16 cells D- 4 cells
24	How many gates can be used to draw this logical equation $(A+B)(B+C'D)$ A- 6 B- 5 C- 4 D- 7
25	Which of the following is also called a Universal Gate : A- AND B- OR C- NOR D- XOR
26	The output of an OR gate is HIGH (1) when A- All inputs are LOW B- Any input is HIGH C- All inputs must be HIGH D- Only one input is LOW
27	A flip-flop is A- Combinational circuit B- Sequential circuit C- Analog circuit D- Power circuit
28	The purpose of simplification is : A- Increase gates B- Reduce hardware C- Increase power D- Increase delay
29	29) A bit is theunit of data in a digital system. A-Biggest B- Smallest C- Basic D-Same
30	How many cells does a 4-variables K map contain : A- 6 cells B- 8 cells C- 16 cells D- 4 cells